LASTNAME_1

Student-ID: 7200001

Name: Lastname_1, Firstname_1

Follow The Below Instructions: .

1.The exam duration is 2 hours.

2.Read all questions carefully before answering.

3.Answer all required questions

Examiner Part Only:

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Examiner Part Only:

Q1: Explain PCA (Principal Component Analysis). [2]

Q2: What is feature selection? [4]

Q3: Explain ensemble learning. [3]

Q1: What is the difference between population and sample? [2]

Q2: Explain mean, median, and mode. [3]

Q3: What is standard deviation, and why is it important? [3]

Q4: A dataset has 40% missing values. How would you handle it? [4]

Q4: What is the Central Limit Theorem? [2]

Q5: Explain Type I and Type II errors. [3]

Q6: What are window functions? [3]

Q7: What is the difference between correlation and causation? [3]

Q8: What are common probability distributions (Normal, Binomial, Poisson)? [3]

Q9: When would you use a t-test instead of a z-test? [3]

Q10: Explain conditional probability. [4]

A AA

B BB

C CC

D DD

Q11: What is the expected value of a random variable? [3]